

SYLLABUS

COMPUTER ARCHITECTURE & ORGANIZATION

Unit I: Basic Structure of Computer: Bus structures, Addressing Methods, and Machine Program Sequencing: Memory Locations, Addresses, Instruction and instruction sequencing, Addressing Modes, Basic I/O Operations.

Unit II: Memory Unit: Basic Concepts, Memory Hierarchy, Semiconductor RAM Memories, Internal Organization of Memory Chips, Static Memories, Dynamic Memories, Read Only Memories, Speed, Size and Cost.

Unit III: Processing Unit: Fundamental Concepts, Execution of a Complete Instruction, Hardwired Control, Performance Consideration, Microprogrammed Control, Microinstructions, Microprogram Sequencing.

Unit IV: I/O Organization: Accessing I/O Devices, Interrupts, Enabling and Disabling Interrupts, Handling Multiple Devices, DMA, I/O Hardware, Standard I/O Interfaces: SCSI.

Unit V: Arithmetic: Number Representations, Design of Fast Adders, Signed Addition and Subtraction, Multiplication of Positive Numbers, Booth Multiplier, Fast Multiplication, Integer Division, Floating Point Numbers and Operations.

Unit VI: Parallel Organization and Pipelining: Parallel Processing, Array Processors, The Structure of General-Purpose Multiple Processors, Symmetric, Multiprocessors, Pipelining: Basic concepts of pipelining, throughput and speedup, pipeline hazards.